



**NATIONAL
SENIOR CERTIFICATE /
NASIONALE
SENIORSERTIFIKAAT**

GRADE/GRAAD 11

NOVEMBER 2023

**TECHNICAL MATHEMATICS P1
MARKING GUIDELINE/
TEGNIJSE WISKUNDE V1
NASIENRIGLYN**

**MARKS /
PUNTE** **150**

Marking codes in CAPS / Nasienkodes in KABV	
A	Accuracy / Akkuraatheid
CA	Consistent Accuracy / Deurlopend Akkuraatheid
M	Method / Metode
R	Rounding / Afronding
NPR	No Penalty for Rounding / Geen penalisering vir afronding
NPU	No Penalty for Units omitted / Geen penalisering vir geen eenhede
S	Simplification / Vereenvoudiging
SF	Substitution in the correct Formula / Vervanging in die korrekte formule
AO	Answer Only / Slegs antwoord

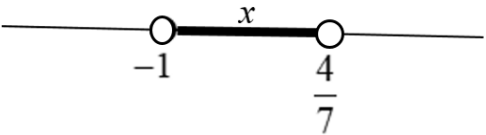
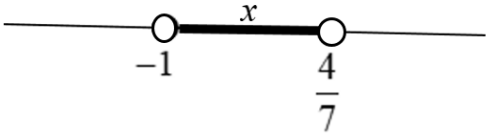
This marking guideline consists of 12 pages /
Hierdie nasienriglyn bestaan uit 12 bladsy

QUESTION / VRAAG 1				
1.1				
1.1.1	$(5\sqrt[3]{3} - x^5)^0 = 1$	✓ 1	A	(1)
1.1.2	$x^{\frac{1}{2}}(3-x)$ $= 3 \cdot x^{\frac{1}{2}} - x^{1+\frac{1}{2}}$ $= 3x^{\frac{1}{2}} - x^{\frac{3}{2}}$	✓ $3x^{\frac{1}{2}}$ ✓ $-x^{\frac{3}{2}}$	A A	(2)
1.1.3	$(\sqrt{3}-3)(\sqrt{3}+3)$ $= (\sqrt{3})^2 - (3)^2$ $= 3-9$ $= -6$	✓ 3 ✓ -9 ✓ -6	A A CA	(3)
1.1.4	$\frac{\log_2 32^{\frac{1}{5}} + \log_2 27^{\frac{1}{3}}}{\log_2 6 + \log_7 x^0}$ $= \frac{\log_2 (2^5)^{\frac{1}{5}} + \log_2 (3^3)^{\frac{1}{3}}}{\log_2 6 + \log_7 1}$ $= \frac{\log_2 2 + \log_2 3}{\log_2 6 + 0}$ $= \frac{\log_2 (2 \times 3)}{\log_2 6}$ $= \frac{\log_2 6}{\log_2 6}$ $= 1$	✓ Prime factors / <i>priemfaktore</i> ✓ $\log_7 1$ ✓ Simplification / <i>vereenvoudiging</i> ✓ Log property / <i>eienskap</i> ✓ Log property / <i>eienskap</i> ✓ 1	A A CA CA CA CA	(6)

1.2.1	$y = \sqrt[x]{\frac{3^{x-1} - 7 \cdot 3^{x+1}}{6 \cdot 9^x}}$ $y = \sqrt[x]{\frac{3^x \cdot 3^{-1} - 7 \cdot 3^x \cdot 3}{6(3^2)^x}}$ $y = \sqrt[x]{\frac{3^x(3^{-1} - 7 \cdot 3)}{6 \cdot 3^{2x}}}$ $y = \sqrt[x]{\frac{\left(\frac{1}{3} - 21\right)}{6 \cdot 3^x}}$ $y = \sqrt[x]{\frac{-62}{6 \cdot 3^x}}$ $y = \left(-\frac{62}{18 \cdot 3^x}\right)^{\frac{1}{x}}$ $y = \frac{1}{3} \cdot \left(-\frac{31}{9}\right)^{\frac{1}{x}}$	✓ Exponential property / <i>eksponensiele eienskap</i> ✓ Prime factors / <i>priemfaktore</i> ✓ Common factor / <i>gemene faktor</i> ✓ Simplification / <i>vereenvoudiging</i> ✓ Simplification / <i>vereenvoudiging</i>	A A CA CA CA	(5)
1.2.2	Even root of a negative number is imaginary and so x must be odd. <i>Ewe wortel van 'n negatiewe getal is imaginêr so x moet onewe wees</i>	✓ Imaginary or non-real / <i>imaginêr of nie-reël</i>	A	(1)
1.3				
1.3.1	$X = 100\,000_2 = 32$ $Y = 111_2 = 7$	✓ $X = 32$ ✓ $Y = 7$	A A	(2)
1.3.2	$X - Y$ $= 100\,000_2$ $\underline{\quad 111_2 \quad}$ $\underline{\quad 110\,01_2 \quad}$ OR / OF $X - Y = 32 - 7 = 25$ $25 = 11001_2$	✓ Method / <i>metode</i> ✓ 11001_2	A A	(2)
1.4				
1.4.1	$\frac{1}{12} \text{ cm}$ $= \frac{1}{12} \times 10 \text{ mm}$ $= 0,8\dot{3} \text{ mm}$	✓ Conversion factor / <i>herleidingsfaktor</i> ✓ $0,8\dot{3}$	A CA	(2)

1.4.2	$= 0,8\dot{3} \text{ mm} = 8,3 \times 10^{-1} \text{ mm}$	✓ 8,3 ✓ 10^{-1}	NPU	CA CA	(2)
					[26]
QUESTION / VRAAG 2					
2.1					
2.1.1	$z = 1$	✓ 1		A	(1)
2.1.2	$x + y - 2z$ $= \log 2 + \log 7 - 2 \log 10$ $= \log (2 \times 7) - \log 10^2$ $= \log 14 - \log 100$ $= \log \left(\frac{14}{100} \right)$ $= \log 0,14$	✓ Substitution / <i>vervanging</i> ✓ Log property / <i>eienskap</i> ✓ Power rule / <i>magreël</i> ✓ S ✓ Log Property / <i>eienskap</i>		A A A CA CA	(5)
2.2					
2.2.1	$3^{x+1} \cdot 3^{x-3} = 1$ $3^{2x-2} = 3^0$ $2x - 2 = 0$ $x = 1$	✓ Exponential prop. / <i>eksponensiele eienskap</i> ✓ 3^0 ✓ Exponential prop. / <i>eksponensiele eienskap</i> ✓ S		A A CA CA	(4)
2.2.2	$\sqrt{48} - x^2 \sqrt{3} = \sqrt{27}$ $\sqrt{2^4 \times 3} - x^2 \sqrt{3} = \sqrt{3^3}$ $4\sqrt{3} - 3\sqrt{3} - x^2 \sqrt{3} = 0$ $\sqrt{3} - x^2 \sqrt{3} = 0$ $\sqrt{3}(1 - x^2) = 0$ $x = 1 \text{ or/of } x = -1$	✓ Prime factors / <i>priemfaktore</i> ✓ Standard form / <i>standaardvorm</i> ✓ S ✓ Factors / <i>faktore</i> ✓ $x = 1$ ✓ $x = -1$		A A CA CA CA CA	(6)
2.2.3	$10^x = 30$ $x = \log 30$ $x = \log (10 \times 3)$ $x = \log 10 + \log 3$ $x = 1 + 0,48$ $x = 1,48$	✓ Logarithm / <i>logaritme</i> ✓ Log property / <i>eienskap</i> ✓ Substitution / <i>vervanging</i> ✓ 1,48		A A CA CA	(4)

2.3	$\begin{aligned} \text{LHS/LK} &= 2^{2x} \cdot 7^{x-1} - 5 \cdot 28^x \\ &= 2^{2x} \cdot 7^x \cdot 7^{-1} - 5 \cdot (2^2 \cdot 7)^x \\ &= 2^{2x} \cdot 7^x \cdot 7^{-1} - 5 \cdot 2^{2x} \cdot 7^x \\ &= 2^{2x} \cdot 7^x (7^{-1} - 5) \\ &= 2^{2x} \cdot 7^x \left(\frac{-34}{7} \right) \\ &= -34 \cdot 2^{2x} \cdot 7^{x-1} \\ &= \text{RHS/RK} \end{aligned}$	✓ Prime factors / <i>priemfaktore</i> ✓ Power rule / <i>magreël</i> ✓ Common factor / <i>gemene faktor</i> ✓ S ✓ S	A CA CA CA CA	(5)
2.4				
2.4.1	$\begin{aligned} C &= \frac{5(F-32)}{9} \\ 9C &= 5(F-32) \\ \frac{9C}{5} + 32 &= F \end{aligned}$	✓ 9C ✓ F subject / <i>onderwerp</i>	A CA	(2)
2.4.2	$\begin{aligned} \frac{9C}{5} + 32 &= F \\ \frac{9(2)}{5} + 32 &= F \\ \therefore F &= 35,6^\circ\text{F} \end{aligned}$	✓ Substitution / <i>vervanging</i> ✓ 35,6	CA CA	(2)
		NPU		[28]

QUESTION / VRAAG 3				
3.1				
3.1.1	$x(x-2)=0$ $x=0$ or/of $x=2$	$\checkmark x=0$ $\checkmark x=2$	A A	(2)
3.1.2	$x^2 + x\left(3 - \frac{5}{x}\right) = 0$ $x^2 + 3x - 5 = 0$ $x = \frac{-(3) \pm \sqrt{(3)^2 - 4(1)(-5)}}{2(1)}$ $\therefore x = 1,2$ or/of $x = -4,2$	$\checkmark$ Standard form/ <i>standaardvorm</i> $\checkmark$ Substitution / <i>vervanging</i> $\checkmark x = 1,2$ $\checkmark x = -4,2$	A CA CA CA	(4)
3.1.3	$-7x^2 - 3x + 4 > 0$ $(-7x+4)(x+1) > 0$ OR / OF $x = \frac{3 \pm \sqrt{(-3)^2 - 4(-7)(4)}}{2(-7)}$ CVs/KWs: -1 and/en $\frac{4}{7}$ $x < -1$ and/en $x > \frac{4}{7}$ OR / OF $-1 < x < \frac{4}{7}$  OR / OF $7x^2 + 3x - 4 < 0$ $(7x-4)(x+1) < 0$ OR / OF $x = \frac{3 \pm \sqrt{(-3)^2 - 4(-7)(4)}}{2(-7)}$ CVs/KWs: -1 and/en $\frac{4}{7}$ $x < -1$ and/en $x > \frac{4}{7}$ OR / OF $-1 < x < \frac{4}{7}$ 	$\checkmark$ Substitution/ <i>Vervanging /</i> <i>Factors / Faktore</i> $\checkmark$ Critical values / <i>kritiese waardes</i> $\checkmark$ Correct notation / <i>korrekte notasie</i> $\checkmark$ Number line / <i>getallelyn</i>	A CA A A OR / OF A CA A A	(4)

3.3				
3.3.1	Radius = 31 mm	✓ 31	A	(1)
3.3.2	$C = 2\pi r$ $C = 2\pi(31) = 62\pi$	✓ 62π	CA	(1)
3.3.3	$V = \pi(31)^2(73)$ $V = 220392.25 \text{ mm}^3$	✓ Substitution / <i>vervanging</i> ✓ Value of / <i>waarde van V</i>	A CA	(2)
3.3.4	$C = 2\pi r$ $r = \frac{C}{2\pi}$ $A = \pi r^2$ $A = \pi \left(\frac{C}{2\pi} \right)^2$ $A = \pi \left(\frac{C^2}{4\pi^2} \right)$ $\therefore A = \frac{C^2}{4\pi}$	✓ r ✓ Substitution / <i>vervanging</i> ✓ Simplification / <i>vereenvoudiging</i>	A CA CA	(3)
3.3.5	$A = \pi r^2$ $5665,36 = \pi r^2$ $r = \sqrt{\frac{5665,36}{\pi}}$ $r = 42,47$ $d = 2r = 84,94 \text{ mm}$	✓ Substitution / <i>vervanging</i> ✓ Value of / <i>waarde van r</i> ✓ d	A CA CA	(3)
				[26]

QUESTION / VRAAG 4				
4.1	Real, Irrational and Unequal / <i>Reëel, irrasionaal en ongelyk</i>	✓ Real / <i>Reëel</i> ✓ Irrational/ <i>irrasionaal</i> ✓ Unequal / <i>ongelyk</i>	A A A	(3)
4.2				
4.2.1	$x = 1$	✓ 1	A	(1)
4.2.2	$\frac{x+1}{x-1} + 2 = 0$ $\frac{x+1}{x-1} = -2$ $x+1 = -2x+2$ $3x = 1$ $\therefore x = \frac{1}{3}$	✓ $\Delta = 0$ ✓ Simplification / <i>vereenvoudiging</i> ✓ Value of / <i>waarde van</i> x	A CA CA	(3)
4.2.3	$\frac{x+1}{x-1} < -2$	✓ Less than / <i>minder as</i> -2	A	(1)
				[8]
QUESTION / VRAAG 5				
	$f(x) = -\frac{2}{x} - 2$ and / <i>en</i> $g(x) = -\sqrt{16-x^2}$			
5.1	$x = 0$ and / <i>en</i> $y = -2$	✓ $x = 0$ ✓ $y = -2$	A A	(2)
5.2	$x = 4$ or/of $x = -4$	✓ $x = 4$ ✓ $x = -4$	A A	(2)
5.3	$f(x) = -\frac{2}{x} - 2$ $0 = -\frac{2}{x} - 2$ $2 = -\frac{2}{x}$ $x = -1$	✓ $y = 0$ ✓ Simplification / <i>vereenvoudiging</i> ✓ $x = -1$	A CA CA	(3)
5.4	$g(0) = -\sqrt{16-(0)^2} = -4$	✓ -4	A	(1)

5.5		f ✓ Shape/ vorm A ✓ x-intrcpt/afsnt CA ✓ Asym / Asimp CA g ✓ Shape/ vorm A ✓ x-intrcpt/afsnt CA ✓ y-intrcpt/afsnt CA	(6)	
5.6	See graph / sien grafiek	✓ All points / alle punte	A	(1)
5.7	$x \neq 0$ OR / OF $x \in (-\infty ; 0) \cup (0 ; \infty)$ OR / OF $x < 0$ or / of $x > 0$ OR / OF $-\infty < x < 0$ or / of $0 < x < \infty$	✓ Solution set / oplossingvers	A	(1)
5.8	$-\sqrt{16-x^2} = -2$ $16-x^2 = 4$ $12 = x^2$ $\therefore x = \pm \sqrt{12}$ Accept/aanvaar: $x = 2\sqrt{3} \approx 3,46$ $-\sqrt{12} < x < \sqrt{12}$ OR / OF $-\sqrt{12} < x$ and/en $x > \sqrt{12}$ OR / OF $x \in (-\sqrt{12} ; \sqrt{12})$	✓ M Equating/ gelykstelling ✓ $x = \pm \sqrt{12}$ ✓ Critical values / kritiese waardes ✓ Correct notation / korrekte notasie	A CA CA A	(4)
				[20]

QUESTION / VRAAG 6				
6.1	$y = -4$	✓ $y = -4$	A	(1)
6.2	B (0 ; -3)	✓ $x = 0$ ✓ $y = -3$	A A	(2)
6.3	$-2 = a^1 - 4$ $-2 + 4 = a$ $\therefore a = 2$	✓ Substitution / <i>vervanging</i> ✓ Simplification / <i>vereenvoudiging</i> ✓ Same exponents rule / <i>dieselfde eskponente reël</i>	A CA CA	(3)
6.4	$y > -4$	✓ $y > -4$	A	(1)
6.5	$0 = 2^x - 4$ $2^2 = 2^x$ $\therefore x = 2$	✓ $f(x) = 0$ ✓ x -value / <i>waarde</i>	A CA	(2)
6.6	K (1; 2)	✓ $x = 1$ ✓ $y = 2$	A A	(2)
				[11]
QUESTION / VRAAG 7				
7.1	A (-1 ; 0) and/ en D (3 ; 0)	x-intercepts only / slegs x-afsnitte: 1/2 ✓ A (-1 ; 0) ✓ D (3 ; 0)	A A	(2)
7.2	B (0 ; 6)	✓ $x = 0$ ✓ $y = 6$	A A	(2)
7.3	Axis of symmetry = $\frac{-1+3}{2}$ <i>Simmetriese – as = 1</i> Maximum turning = $f(1) = (6 - 2)(1 + 1)$ <i>Maksimum draaipt = 8</i>	✓ Method / <i>Metode</i> ✓ $x = 1$ ✓ 8	A CA CA	(3)
7.4	$x = 0$ and/en $x = 3$	✓ $x = 0$ ✓ $x = 3$	CA CA	(2)
7.5	$h(x) = (6 - 2(x - 2))(x - 2 + 1)$ $h(x) = (6 - 2x + 4)(x - 1)$ $h(x) = (10 - 2x)(x - 1)$	✓ Substitution / <i>vervanging</i> ✓ Simplification / <i>vereenvoudiging</i>	A CA	(2)
7.6	Area of $\Delta BOD = \frac{1}{2}(6)(3) = 9$	✓ SF ✓ Area = 9	A CA	(2)
				[13]

QUESTION / VRAAG 8				
8.1	$7,8\% \text{ of/van } R2\ 567 = \frac{7,8}{100} \times R2\ 567$ $= R200,23$	✓ R200,23	A	(1)
8.2	$i_{eff} = \left(1 + \frac{i}{m}\right)^m - 1$ $i_{eff} = \left(1 + \frac{0,078}{12}\right)^{12} - 1$ $i_{eff} = 8,01\%$	✓ Formula / formule ✓ Substitution / vervanging ✓ 8,01%	A CA CA	(3)
8.3	$A = P(1-i)^n$ $R800\ 000 = P\left(1 - \frac{0,072}{4}\right)^{6 \times 4}$ $P = \frac{R800\ 000}{\left(1 - \frac{0,072}{4}\right)^{6 \times 4}}$ $P = R1\ 237\ 126,99$	✓ Formula / formule ✓ i and n ✓ Substitution / vervanging ✓ P the subject / die onderwerp ✓ S	A A A CA CA	(5)
8.4				
8.4.1	$A_1 = R320\ 000(1+0,05)^3$ $A_1 = R370\ 440$ $A_3 = R370\ 440 + R400\ 000$ $A_3 = R770\ 440$	✓ Formula / formule ✓ Substitution / vervanging ✓ M + R400 000	A A CA	(3)
8.4.2	$R950\ 000 = R770\ 440(1+0,058)^n$ $\frac{R950\ 000}{R770\ 440} = 1,058^n$ $n = \log_{1,058}\left(\frac{R950\ 000}{R770\ 440}\right)$ $\therefore n \approx 3,72 \text{ years/jare}$ Period = $3+3,72 \approx 6,72 \text{ years / jare}$ $\therefore$ It will take 7 years	✓ Substitution / vervanging ✓ Simplification / vereenvoudiging ✓ Logarithm / logaritme ✓ $n = 3,72 \text{ years/ jare}$ ✓ Period = 7 years / jare	CA CA A CA CA	(5)
				[16]
TOTAL/TOTAAL:				150